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Back to the Future: 'Yesterday' (and 'Tomorrow') in Indo-European

1. The Proto-Indo-European word for 'yesterday', with widely-attested reflexes,[1] poses notorious difficulties of reconstruction. A thorough critique of previous hypotheses might fill a short monograph, and cannot be undertaken here. Fortunately, two convenient substitutes are available: G.E. Dunkel (2004:127f.) has recently surveyed the most important approaches to the problem, including that of J. Puhvel (1987 = 2002:43-46), who provides more extensive coverage of earlier literature.[2]

2. For Puhvel (1987:317), PIE 'yesterday' is built on the word for 'day' seen in Ved. *áhar-/áhan-* and its Iranian counterparts (EWAia I.154), which belongs, as generally assumed—and despite the severe difficulty raised by the divergent initials—with the word for 'day' in Germanic (Goth. *dags*, etc.). According to Puhvel, the resulting PIE **dʰógʰos* or **dʰógʰr̥* 'day' would have had an oppositional form **(dʰ)ĝʰ(i)i̯ós* [sic], with the "contrastive" suffix **-i̯os-* familiar from primary comparative formations,[3] which in its contrastive usage signals the marked member of a binary opposition. The suffixless locative **(dʰ)ĝʰ(i)i̯és* of this form **(dʰ)ĝʰ(i)i̯ós* would have denoted, in Puhvel's words, "the one of two adjacent days which contrasted with 'now', thus 'the other day'"; the meaning 'yesterday' would be a "conventional" designation, "due in large measure to the pre-emption of futurity by the 'morrow' words" (referring to expressions for 'tomorrow' derived from words for 'morning',

[1] Among others: Ved. *hyás*, Gk. (ἐ)χθές, Lat. *herī*, OIr. *in-dé*, Alb. *dje* (see IEW 416, LIPP s.v. **ĝʰdʰi̯és*, Mallory and Adams 1997:654, 2006:301), plus derived forms (Lat. *hesternus* 'yesterday's', Gk. χθιζός 'id.', etc.). On the British Celtic forms (MW *doe* etc.), see recently Schrijver (2007:315).

[2] Below, I cite Puhvel's article in the 1987 pagination, also provided in the 2002 reprint.

[3] With original amphikinetic inflection (Ringe 2006:63); thus, instead of Puhvel's anomalous "**(dʰ)ĝʰ(i)i̯ós*," one would expect rather (animate) nom. **dʰégʰ-i̯ōs*, acc. **dʰégʰ-i̯os-m̥*, obl. **dʰĝʰ-is-*´, loc.sg. **dʰĝʰ-(i)i̯és(+i)*, neut. nom./ acc. **dʰégʰ-i̯os* or **dʰégʰ-is*.

1987:315; more on this point below, §6). Morphologically, according to Puhvel, the entire construct would be "quite parallel to Gk. κῦδος or *κῦδαρ : *κύδιος."

This ingenious theory (prefigured in various ways by Specht, Brandenstein, and Durante, as Puhvel notes) seems to have found adherents,[4] and at first sight the reconstruction *$(d^h)\hat{g}^h(i)\underline{i}\acute{e}s$ seems at least to provide a good formal match to the attested IE forms, always assuming (as with earlier theories) diverse cluster-reductions in different IE areas.[5] In one respect, however, the formal match is less than ideal, as Dunkel has pointed out (2004:128): if one accepts the attractive suggestion that Tocharian A *ksär* 'early, in the morning' (see §6 below on the semantics) continues a form of the PIE word for 'yesterday',[6] this casts doubt on Puhvel's reconstruction, since an initial sequence *$d^h\hat{g}^h$- should remain intact in Tocharian (cf. TA *tkaṃ* beside Gk. χθών, to PIE *$d^h\acute{e}\hat{g}^h$-*ōm* 'earth', both forms showing a generalized zero-grade root vocalism). In contrast, "classical" reconstructions of 'yesterday' with PIE thorn notations, of the type *$\hat{g}^h\underline{p}i\acute{e}s$ (cf. IEW 416), face no such difficulty.

Puhvel's *$(d^h)\hat{g}^h(i)\underline{i}\acute{e}s$ is equally problematic from several non-phonological perspectives. First, it is questionable whether the primary comparative (or, if one prefers, "contrastive") suffix could have functioned with the "root" in question in the way Puhvel describes. As a primary formation based on adjectival roots (or the participial/adjectival value derivable from verbal roots),[7] *-*i̯os*- in a form *$(d^h)\hat{g}^h(i)\underline{i}\acute{e}s$ '(on) the one of two days contrasting with now' would be inherently anomalous: there is no "root" †*$d^he\hat{g}^h$-, with appropriate adjectival meaning, corresponding to the nominal prime for 'day' that Puhvel wishes to reconstruct (on which see further below), and no evidence that *-*i̯os*- at some remote stage could be used with nominal primes at all, much less in the contrastive meaning Puhvel assigns to it.[8] Note that a form like Lat. *peior* 'worse', which superficially appears to be based on **ped*- 'foot', offers no support: as is generally acknowledged (e.g. IEW 791; Leumann 1977:496; Watkins 2000:62), the comparative suffix here is based on the participial (hence adjectival) sense 'stumbling' derivable from the verbal root **ped*- 'walk, fall' (Ved. *pádyate* etc.), essentially as in the Vedic type *yájīyas*- 'sacrificing better'; nor is it attractive to think in terms of 'the one of two contrasting feet (i.e. the left foot??)'.[9]

Secondly, Puhvel's comparison to "Gk. κῦδος or *κῦδαρ : *κύδιος" provides, at best, a "parallel" of only the most limited sort. To begin with, a form *κῦδαρ—the putative parallel to IIr. **áhar*- —does not exist, and there is little reason to reconstruct such an entity.[10] Insofar as there is a parallel between *κύδιος (the

[4] Apparently following Puhvel: Mallory and Adams (1997:654, 2006:301), also with *$(d^h)\hat{g}^h\underline{i}es$ for PIE 'yesterday', though without internal segmentation or further explanation of the formation; similarly Watkins (2000:20), with pre-form *$d^h\hat{g}^h(\underline{i})es$-, where the bracketed glide and the hyphen presumably reflect the focus of that work on Germanic. But see Puhvel (1987:316) on previous scholars operating with a general formula of the type *$(d^h)\hat{g}^h(\underline{i})es$.

[5] On the consonantal treatments, attributable at least in part to sandhi conditions, see Mayrhofer (1986:155, 167-168).

[6] See especially Hilmarsson (1991); earlier references in LIPP s.v. *$\hat{g}^hd^h\underline{i}\acute{e}s$.

[7] As Dunkel points out (LIPP s.v. **pro*), Lat. *prior* 'earlier' (< **pri-i̯os*-, built to the allomorph **pri* of *pro* 'before') anomalously shows *-*i̯os*- based on a particle, but this may be an Italic innovation.

[8] On the meaning of *-*i̯os*-, see recently Ringe (2006:63): an original "elative" formation, with a sense like 'exceptionally X' (X = adjectival root). I use a dagger to signify a questionable reconstruction or gloss.

[9] Some other attempts (whether correct or not) to identify contrastive *-*i̯os*- in isolated forms are at least properly based on roots with adjectival (or participial) meaning: e.g. Rasmussen (1986:304-05) on Gk. σταῖς 'dough made with wheat flour' (to the root of Ved. *styā́yate* 'coagulates, stiffens', thus 'dough' < *'becoming more compact'), and on PIE *$\check{a}\underline{u}is(-)$, i.e. Ved. *āvíḥ* 'plainly' etc. (to the root of Hitt. *ūḫḫi/aušzi* 'sees', thus 'plainly' < *'more perceptibly').

[10] No longer acceptable is the *κῦδαρ that Benveniste (1935:17) took to be presupposed by Hom. κυδρός 'glorious', κυδάλιμος 'id.', and κυδαίνω 'glorify'. The gloss-word κύδαρ· τάφος (Hsch.) does not belong here, especially if its assignment to Macedonian is correct (see e.g. Schwyzer 1939:71).

presumed primary comparative behind the synchronic comparative κυδίων [Euripides+], cf. superlative κύδιστος [Hom.+]) and κύδος on the one hand, and contrastive **(d^h)ĝ^h(i)i̯és* beside nominal **d^hóĝ^hos* on the other, the similarity would be purely "typological." In other words, there is indeed good evidence for a pattern, already well-established in Homer, whereby primary comparatives (with pendant superlatives) could be built to *s*-stem abstracts: thus Hom. κέρδιον/κέρδιστος to κέρδος 'profit', ἀλγίων/ἄλγιστος to ἄλγος 'grief', ῥίγιον/ῥίγιστος to ῥῖγος 'cold', among other such forms (Risch 1974:89; Chantraine 1973:112). But even if this is the correct interpretation of (post-Homeric) κυδίων/Hom.+ κύδιστος beside κῦδος (more on this point directly), this says nothing about the status of such a derivational process for PIE, especially one involving the primordial contrastive sense of the suffix Puhvel assumes for **(d^h)ĝ^h(i)i̯és*, versus the synchronically productive comparative sense in Hom. κέρδιον, etc. ('profit' >> 'more profitable', etc., not †'one of two contrasting profits'). It is likely, moreover, that the background of κῦδος (together with its cognate OCS *čudo* 'marvel') is to be conceived of differently in any case. As a form belonging to the Caland system,[11] κῦδος (and OCS *čudo*) could be interpreted as a secondary abstract based on an inherently adjectival or verbal root ('[magically] powerful' or 'having [magic] power' vel sim.), as with many abstracts beside basic Caland adjectives, of the type μῆκος 'length' beside μακρός 'long' (comparative μάσσων, superlative μήκιστος). Even more clearly secondary are cases with zero-grade *s*-stem, of the type κράτος 'might' (: κρατύς/ κρατερός 'mighty'), βάρος 'weight' (: βαρύς 'heavy'), τάχος 'speed' (: ταχύς 'swift'), and others. On the general pattern, see Nussbaum (1986:141), who includes κῦδος among such forms, given its apparent zero-grade. According to an alternative theory about the root of κῦδος (see the discussion, with further references, by de Lamberterie 1996), the zero-grade is only apparent: the form somehow replaces an ordinary *eu*-diphthong, with the acute intonation of Sl. **čŭdo* (traditionally interpreted as from **kéud-es-*) explained via Winter's Law, and not as an old lengthened grade or a form with laryngeal-final root. In that event, the profile of κῦδος (: κύδιστος : κυδρός) would be much like that of Hom. κῆδος 'care' (also with likely inherited *s*-stem, cf. Gothic neut. *hatis* 'anger') beside κήδιστος 'most beloved', with *ro*-stem in Av. *sādra-* 'sorrow'. Here, however, the basis in a verbal root is clear, cf. Gk. κήδομαι 'be distressed' and LIV s.v. **ḱeh₂d-*; and see above on the possibility of a verbal or adjectival root as the basis for κῦδος. In summary, then, the backgrounds of neither "*κύδιος" nor κῦδος offer effective parallels for Puhvel's conception of a contrastive **(d^h)ĝ^h(i)i̯és* beside a nominal prime **d^hóĝ^hos*.

Finally, one is reluctant to believe that the widely-distributed PIE word for 'yesterday' is based not only on the most poorly-attested and problematic of the words for 'day' (as opposed to the widespread forms based on the root noun **dei̯-/*di-*; Rau 2006; LIPP s.v. **dei̯-*), but specifically on the form of that word found in Germanic, where the initial dental, as has always been thought, is more easily taken as secondary (via sandhi effects or contamination) than the absence of that element in the cognate Indo-Iranian forms. For the latter, Puhvel can point only to the "movable dental" of forms like Ved. *áśru-* 'tear' beside Gk. δάκρυ 'id.'—a clear case of *obscurum per obscurius*.

3. Equally ingenious—but likewise difficult to credit—is Dunkel's own theory, offered in a study he himself characterized as "unabashedly speculative" (2004:117). This approach (2004:128, cf. LIPP s.vv. **ĝ^hd^hi̯és* and *?***es*) operates with a (triply-starred, thus presumably "pre-pre-PIE") particle ****es*, with "resultative" meaning—an entity that is claimed to account, among other things, for the PIE augment, *s*-mobile, the *s*-aorist, various *s*-enlarged roots, (one source of) nominal *es*-stems, the personal ending of the zero-person (Hitt. *daiš*, Phryg.

[11] Cf. especially Hom. κυδι- (e.g. κυδιάνειρα 'having famous men', κύδιμος 'glorious'), κυδρός 'glorious'; basic discussion in Risch (1974:65).

εδαες, etc.), *-s* in other personal endings (e.g. 3pl.perf. **-er-s*), insular Celtic main-clause enclitic **(e)s*, and PIE 'yesterday'. For the last item, moreover, Dunkel's proposal crucially involves a supposed (pre-pre-?)PIE instance of homoeoteleutic perseveration, or, in an alternative scenario, contamination. Dunkel begins with the assumption that the basic word for yesterday was simply **ĝʰés* (cf. Lat. *her-ī*). A further extension via the locative adverbial particle **-dʰí* (Gk. οἴκοθι 'at home', etc.) yields **ĝʰz-dʰí*; then, perseveration based on resultative contexts with **es* earlier in the phrase or sentence (**es* ... **ĝʰzdʰí*) yields a new form (**es* ...) **ĝʰzdʰi̯+és*, whence (with the usual consonantal reductions) the attested forms. Alternatively, a form **ĝʰzdʰi̯és* could have arisen by contamination of **ĝʰz-dʰí* with the original basic form **ĝʰés*.

Among many questions raised by this proposal is the nature of the odd-looking form **ĝʰés* 'yesterday', about which Dunkel offers the following conjecture: "Perhaps even the simple **ĝʰes* arose from **ĝʰŏ* 'behind' by the same perseveration" (2004:128). But this hardly constitutes a satisfactory account of an entity like **ĝʰés*.[12] As we shall see, however, the particle **ĝʰō* 'behind' (as well as an insight of Dunkel's about the background of that particle) may well provide a key to a better understanding of the problems surrounding 'yesterday'.

Dunkel's solution, in short, "works," insofar as the pieces of the puzzle fit together, *grosso modo*. But an "unabashedly speculative" account of this kind must be valued less highly than an analysis with fewer special assumptions, and based more closely on actual PIE (as opposed to presumed pre-pre-PIE) parallels.

4. Just such an analysis of the IE 'yesterday' material is due to J. Schindler (1977:34), who posited (as a reinterpretation of the traditional **ĝʰi̯és* cited in §2) a PIE compound **ĝʰ-di̯es*, "consisting of the zero grade of an adverb or a deictic particle plus a case form of a word for 'day' which would also appear in Ved. *sa-dyáḥ* 'within one day'. A different case form is found in Ved. *a-dyá* 'today' from **e-di̯é*." This approach is by now widely accepted: in addition to Mayrhofer (1986, n.5 above), see e.g. EWAia II.822 (s.v. *hyás*), Sihler (1995:225-26), Meiser (1998:97), Hackstein (2005:172), Rau (2006), among many others. Even Dunkel regards this theory as a plausible solution (LIPP s.v. **ĝʰdʰi̯és*), and considers the comparison with Ved. *sadyás* and *adyá* (to which he adds Lat. *hodiē*, Fal. *foied*) to be "attractive" (2004:128), as indeed it is.

Dunkel's reservations, however, are important. The etymology and meaning of the particle in question remain unknown, and the idea that it may have been "deictic" (as conceived earlier by Brugmann and others) is particularly problematic, on at least two counts (Dunkel 2004:127-28). First is "Brugmann's utterly unwarranted assumption that the initial **ĝʰ-* is near-deictic in function," since this "deictic use of **ĝʰo/i* is an innovation of Latin-Faliscan only, not a proto-linguistic feature" (see also LIPP s.v. †**ĝʰo-*); "[n]or," secondly, "is a zero-grade of this particle otherwise known." To this it may be added that evidence for a *far-deictic* pronoun or particle of the requisite form, as implied by the glosses of some scholars who follow Schindler's formal analysis (thus e.g. Meiser 1998:97, "quasi 'an jenem Tag'"), is likewise lacking. In the remainder of this paper, I propose to address these problems by developing two suggestions already put forward, one by Schindler himself, and another by Dunkel.

5. Although it does not appear in the published version of his analysis of PIE 'yesterday', Schindler regularly appended an additional suggestion to his basic hypothesis, in his lectures from the same period.[13] Referring to the particle **ĝʰ-* in his reconstruction, he added that some confirmation for its form and meaning might come from a possible comparison with the

[12] In contrast, e.g. **ḱas-* 'series, row' (whence various adverbial forms) has a plausible basis in a root noun (see Dunkel 2007:54-55, following Klingenschmitt 1975), and **pr-es-* (as in Gk. πρέσβυς 'elder') can be interpreted as a genitive-ablative form (García Ramón 1985).

[13] I refer to Schindler's lectures on Indo-European phonology and noun morphology during the 1970's (Harvard University, Department of Linguistics).

*$\hat{g}^h\bar{o}$ 'back, behind' seen in Slavic (OCS *za*).[14] Given the status of this word as an original adverbial particle, the comparison with Ved. *sadyás* is immediately appealing, both formally and semantically: *$\hat{g}^h\bar{o}$'back, behind' + *-di̯-és* '(on the) day' (gen. of time) = 'yesterday' (i.e., as a first approximation, 'on the "behind-day"', as it were), like *$sm̥$- 'one and the same' + *-di̯-és* = 'on the same day'.

There remains, however, the serious formal obstacle of deriving a zero-grade *$\hat{g}^h$- from what appears to be a lengthened-grade *$\hat{g}^h\bar{o}$, for which no other form is attested.[15] But we may attempt to remove this difficulty by invoking a suggestion of Dunkel (LIPP s.v. ?*$\hat{g}^h\bar{o}$), who notes that the semantics of the particle *$\hat{g}^h\bar{o}$ might favor an etymological connection with the verbal root *$\hat{g}^heh_1$- 'zurücklassen' (see LIV s.v.; Ved. *jáhāti* etc.), thus implying in fact *$\hat{g}^hoh_1$.[16] The immediate formal result of this reconstruction (the semantic implications of which are explored in the next section) is that the word for 'yesterday' would have shown this first member in its zero-grade form *$\hat{g}^hh_1$-, given the accented ending: i.e. *$\hat{g}^hh_1$-*di̯-és* (cf. *$sm̥$-*di̯-és* > Ved. *sadyás*).[17] There is then but a short step between *$\hat{g}^hh_1$-*di̯-és* and Schindler's *$\hat{g}^h$-*di̯-és*, since the loss of the laryngeal in this context can be motivated on independent grounds, according to the general pattern **CH.CC* > **C.CC* (Schmidt 1973; Hackstein 2002).[18] Although this pattern is best attested for sequences with the laryngeal and preceding consonant in the coda position of syllables (thus **VCH.CC* as input; cf. *$u̯erh_1.d^hh_1o$-*, n.18), this does not pose serious difficulties: either the process in question should be generalized to include syllable-onset (here word-initial) sequences as in *$\hat{g}^hh_1$-*di̯-és* (see Mayrhofer 1986:138 on *ph_2tr-*éi̯*), or else the reduction process may also have operated under sandhi conditions, as is assumed anyway for the further reductions of initial clusters in the daughter languages (cf. Mayrhofer, n.5 above). The fact that this reduction occurred at a very early period, finally, is evident from the data: thus, for the example used here, note the "aniṭ" forms for 'word' to be reconstructed already for PIE: Lith. *vārdas* (not †*várdas*), Lat. *uerbum* (not †*ueridum* or †*uerdum*).

6. A longstanding problem with the etymon for 'yesterday' is that some of its forms mean 'tomorrow', principally Goth. *gistra-* (in *gistra-dagis*, Mt. 6:30) and other Germanic material, with these perhaps comparable to TA *ksär* 'in the morning, early' (< *'tomorrow'?),

[14] Cf. Dunkel's own hunch (noted above, §3) that his etymon *$\hat{g}^hés$ 'yesterday' might somehow belong with the same particle *$\hat{g}^h\bar{o}$. Note Schindler's published phrasing (cited in §4, here with emphasis added): "**an adverb** or a deictic particle."

[15] Beside Sl. *za* (and perhaps the noun *zadŭ* 'backside'): also the E.Lith. preposition *ažù, až* 'behind, beyond' (from earlier *ažuo* < proximate *$o\hat{g}^h\bar{o}$, ore on which directly below, n.16), and, if it belongs here at all, the Armenian proclitic preposition and accusative marker *z-* (also *zow-* in *zow-art'* 'sober, prudent'), see Dunkel, LIPP s.v. ?*$\hat{g}^h\bar{o}$ (with further references).

[16] Similarly Klingenschmitt (1982:138, n.3), who writes "*$\hat{g}^hoh$," although this may be a reflexive rewriting of the long vowel. A potential obstacle to this reconstruction would appear to be the *$o\hat{g}^h\bar{o}$ of Baltic (n.15); and indeed, as Dunkel observes (LIPP s.v. ?*$\hat{g}^h\bar{o}$), the alternation between *$o\hat{g}^h\bar{o}$ and *$\hat{g}^h\bar{o}$ has never been satisfactorily explained. But an attractive solution would be to posit, for Baltic, a conjoined *h_2o $\hat{g}^h\bar{o}$ 'to the back', with the particle *h_2o 'bei, neben, zu' (see LIPP s.v. 1. *h_2o), in a collocation much like *h_2o-b^hi 'to here' (Ved. *abhí*, OCS *ob(ĭ)*) or *h_2o b^ho- 'both' (OCS *oba*, Lith. *abù*) ~ *h_2u b^ho- (Ved. *ubhá-*, GAv. *uba-*).

[17] On the accentual pattern: similarly e.g. imperative *h_1i-$d^hí$ 'go!' vs. *$h_1éi̯$-Ø; Dunkel (2004:127), Rix (1992:264).

[18] See also Vine (2006:504), with further references, on the well-attested subtype *$CH.d^hh_1o$- > *$C.d^hh_1o$-: e.g. *$u̯erh_1.d^hh_1o$- 'utterance-making' (cf. *$u̯erh_1$- 'speak') > *$u̯erd^h(h_1)o$- 'word', etc. The process may in fact have been favored for laryngeals in pretonic syllables (as in *$\hat{g}^hh_1$-*di̯-és*): see Mayrhofer (1986:138) and EWAia II.673, s.v. *ślakṣṇá-*. Werner Knobl (Kyoto University) points out to me that the loss of laryngeal in a form like *$\hat{g}^hh_1$-*di̯-és* could in principle be assigned to the rubric of laryngeal loss in end-stressed compounds (e.g. Mayrhofer 1986:125, 140). But the precise parameters of this process (which itself is somewhat controversial) remain unclear, and so the laryngeal loss here has provisionally been assigned to the process whose structural description seems to match the form most closely.

already cited (§2) in a different connection.[19] As Puhvel notes in his own discussion of the problem (1987:317), the traces of this meaning have generally been "desperately explained away" (see the references he cites to Brugmann and Strunk); and indeed, the ability of Puhvel's theory to account for this sense in a natural way is its strongest feature. Thus, addressing the Germanic material, and with the background of widespread IE formations for 'tomorrow' based on expressions for 'morrow, morning', Puhvel shows how the sense 'yesterday', for his original contrastive term 'the one of two adjacent days contrasting with now', would naturally have predominated, while traces of the other predictable meaning (i.e., 'the next day') might also occur.

Yet this strength of Puhvel's theory cannot offset its formal weaknesses (§2); and in any event, this feature of PIE 'yesterday' can also be explained by the reconstruction *$\hat{g}^h h_1$-*di̯-és* based on the *$\hat{g}^h o h_1$ of Sl. *za* and its cognate forms. Conventional glosses like 'back, behind', used provisionally above, obscure the fact that this particle is in essence a "shifter" (in Jakobson's still-useful terminology, 1957/1971): its spatial (and temporal) meaning—basically 'on the far side of'—shifts, depending on point of view, much as with "deictic" particles *sensu stricto*. Thus, in OCS *idi za mę, Sotono* (Mt. 4:10),[20] Satan is being asked, from my point of view, to 'get behind me'; but from Satan's point of view, if he is facing me, this movement to the back of me would bring him forward, 'to the other side of me', i.e. 'beyond me' or 'past me': cf. the Russian folk-tale formula *za gorami, za lesami* 'beyond the mountains, beyond the forests'. It is this "prospective" usage—looking *forward* ('beyond') to get to the back of something—that helps account for the causal meaning ('because of', 'for the sake of', in OCS with both accusative and genitive). Likewise, the concomitant notion of reciprocity seems related to the meaning 'in exchange for', 'on behalf of' (in OCS, as in modern Russian, with accusative: *oko za oko, zǫbŭ za zǫbŭ* 'an eye for an eye, a tooth for a tooth', Mt. 5:38). Various temporal usages (with both accusative and genitive) can be understood on the same basis: OCS *za tri lěta* 'during three years', *za rana* 'early in the morning', *za utro* 'id.' (literally 'beyond the morning') > Russ. *zavtra* 'tomorrow'. (The last item is particularly interesting in the present connection.) Given the prevalence of terms for 'tomorrow' based on words for 'morrow, morning', observed by Puhvel, we can readily apply his "preemption" analysis to *$\hat{g}^h h_1$-*di̯-és* '(during) the day that is on the other side': the term will generally have meant 'yesterday', but in principle could also have meant '(on) the day beyond (today)', i.e. 'the next day, tomorrow', a usage that may survive in Germanic and Tocharian (with a shift in the latter to 'in the morning', 'early in the morning').

7. The above conception of the meaning of *$\hat{g}^h o h_1$ 'back, beyond', together with Dunkel's hypothesized connection with *$\hat{g}^h e h_1$- 'zurücklassen' (§5), may help clarify certain problems with the form and meaning of this verbal root, and at least one other that may be related to it.

It is convenient to begin by recalling the LIV treatment of relevant material, distributed among lemmata for three distinct PIE roots: *$\hat{g}^h e h_1$- 'zurücklassen', ?*$\hat{g}^h eH$- 'die Körperstellung verändern, sich bewegen' (note the question mark, more on which shortly), and *$g^h eh_1$- 'kommen, erreichen'. This conception owes much to a treatment by Klingenschmitt of Arm. *gam* 'come' and forms said to be related to it (1982:85-86); and indeed, Arm. *gam* seems to require a plain velar *g^h-, whereas Ved. *jáhāti* 'leaves behind' and Av. *zazāiti* 'leaves behind; is victorious' call for a palatal *$\hat{g}^h$-, hence (in part) *$g^h eh_1$- vs. *$\hat{g}^h eh_1$-. The remaining forms, however, have been assigned to these roots on the basis of two other criteria, namely (a) semantics, and (b) the theory of the morphology of reduplicated presents under which LIV operates.

[19] For the Gothic form, see e.g. Mallory and Adams (1997:654): "The etymological basis of Goth. *gistradagis* is puzzling; we have no PIE word for 'tomorrow'"; see also Hackstein (2005:172) for a discussion of the Tocharian form (together with the Germanic material) as a possible semantic archaism.

[20] Also with instrumental (*za mŭnojǫ*, Mt. 16:23, Mk. 8:33, Lu. 4:8). For a convenient survey of the usage of *za* in OCS and in modern Slavic languages, see Vaillant (1977:145-46); more detail on the OCS case usage in Xodova (1971:54, 86-92, 146-58, 171-72).

Under ?*$\hat{g}^h eH$-, we find only Ved. *jíhīte* 'moves', a corresponding Avestan form, and a middle *s*-aorist (Ved. *ahāsata*). Traditionally, however, *jíhīte* has been viewed (see EWAia II.813-14, with references) simply as the middle corresponding to *jáhāti*,[21] with secondary *i*-reduplication, a feature LIV in fact prefers (s.v. ?*$\hat{g}^h eH$-, n.3; similarly Jasanoff 2003:129, though with a different conception). LIV comes close to conceding the traditional approach (s.v. ?*$\hat{g}^h eH$-, n.1): "Vorliegende Wurzel unterscheidet sich formal fast durch die mediale Diathese von *$\hat{g}^h eh_1$- 'zurücklassen'; eine Vereinigung beider Wurzeln bedürfte genauerer semantischer Untermauerung." As for *$g^h eh_1$-: apart from the apparent need for a plain velar occasioned by Arm. *gam* (cf. above), LIV places here two other sets of intransitive forms, namely Gmc. *$g\bar{e}$- 'go away, go' (OE *gān* etc.; see Mottausch 1997) and the forms of Gk. κιχάνω 'reach, attain', even though both of these are phonologically ambiguous between palatal and velar initials. In addition to the intransitive meaning of these forms, this assignment is governed by the *i*-reduplication and *e*-grade root in Greek (as in 3du.aor. κιχήτην, interpreted—again following Klingenschmitt (1982:86)—as an old reduplicated present integrated into the aorist system), and the *e*-grade root in Gmc. *$g\bar{e}$-, since LIV operates (p. 16) with a systematic PIE distinction between *i*-reduplication with *e*-grade root vs. *e*-reduplication with *o*-grade root, for the strong forms of athematic (active) reduplicated presents. If not for this fact, then, as LIV admits (s.v. *$g^h eh_1$-, n.3), Gmc. *$g\bar{e}$- —which, as a dereduplicated form (like Arm. *gam*), lacks all surface trace of reduplication—might belong instead with LIV's *$\hat{g}^h eH$- (also an intransitive verb of motion).

Yet this alleged PIE distinction rests on very little, as Jasanoff has discussed with regard to both the root vocalism (2003:67 with n.8) and the reduplicating vowel (2003:129-30, 132 with n.12) of reduplicated presents. The way is open, then, for a synthesis of this verbal material, which seems desirable in any case, given the considerable formal and semantic overlap. A step in this direction has recently been taken by Haudry (2005), who offers a twist on Hoffmann's interpretation of Av. *zā-/zaz-* 'is victorious' < *'leaves behind (the opponents in a race)' (Hoffmann 1968:284-85 = 1975:223-24), applied to the Greek material: Haudry envisages κιχάνω 'reach, attain' in terms of the other perspective, i.e. that of 'reaching' the finish line or the turning point, which is then 'left behind'. Haudry points, however, to one of the most puzzling features of this material, namely the discrepant diatheses: the transitive construction of κιχάνω (generally with direct object, e.g. τέλος 'end, goal') differs from the Avestan "intransitivization" (via suppression of the direct object 'opponents'), and seems to match the construction of Ved. *jáhāti* 'leaves behind', which in turn differs from intransitive *jíhīte* and Gmc. *$g\bar{e}$-. Mottausch, similarly, draws attention to the formal and semantic conundrum posed by the Germanic material (1997:260), where the form (assuming dereduplication) matches that of Ved. *jáhāti*/ Av. *zazāiti*, but the intransitive meaning recalls *jíhīte*. He then proposes an elaborate scenario (1997:261-68), according to which Proto-Germanic inherited both active and middle forms, but the middle, after having adopted the vocalism of the active, survived as a deponent after the active had been lost at an early period, and then (with the general loss of deponents) developed as an active verb.

8. As a way of addressing the difficulties (and reducing some of the complexities) just sketched, I suggest that the material assigned by LIV to the roots *$\hat{g}^h eh_1$-, *$\hat{g}^h eH$-, and *$g^h eh_1$-, with the exception of Arm. *gam*, can be aligned in a straightforward way with the particle *$\hat{g}^h oh_1$ 'back, beyond', thus following up and extending Dunkel's suggestion regarding the possible relationship between "*$\hat{g}^h\bar{o}$" and (verbal) *$\hat{g}^h eh_1$-. This can be accomplished by an appeal to what Dunkel calls *radices departiculativae* (PIE verbal roots that originate in particles), a theoretical construct he has recently discussed in some detail (2007:57-59).

For an uninflected language like English, the process known as (part-of-speech) "conversion" includes the development *adverbial particle* >> *verb*, as in examples like those Dunkel cites ("to *up* the ante", "to *down* a drink"), and many

[21] Which itself has an active *s*-aorist (Ved. *ahās*, etc.).

others. Likewise well-attested are similar developments for the older Indo-European languages, for which Dunkel offers a number of instances: among others, e.g. Hitt. *ūpzi* 'rises' (cf. PIE **úp* 'up'), *parai* 'appears, comes/goes forth' (cf. *parā* 'forth, out'; Melchert [forthcoming]), Hom. νοσφίζομαι 'remove oneself, withdraw' (cf. νόσφι 'apart from'), Lat. *iterāre* 'repeat' (cf. *iterum* 'again'), etc. It is entirely to be expected, as Dunkel points out, that similar behavior should have occurred within PIE itself. Thus, as I suggest elsewhere (see for the moment the notice *apud* Dunkel 2007:58), PIE **k̂ei-* 'lie'—which, in its Hittite, Vedic Sanskrit, and Homeric Greek avatars, prominently means 'is [lying] here, is located [in plain sight]' (as opposed to 'is supine')—may have its origin as a "verbalization" or "morphologization" (Dunkel's term) of the near-deictic particle **k̂e/*k̂i* 'here [in front of the speaker]'. Dunkel provides nearly a dozen additional examples for PIE, some of which have long been assumed (e.g. **du-s-* [pejorative prefix] >> **deus-* 'need, lack' [**déus-e/o-* in Gk. δεύομαι 'be lacking']), or which seem particularly attractive (e.g. **pér* 'through' >> **per-* 'go through, traverse' [e.g. **pór-e* in Goth. *faran* 'go, travel', **pér-i̯e/o-* in Gk. πείρω 'pierce', **pi-pér-ti* in Ved. *píparti* 'brings across']). As Dunkel observes (2007:58), "[t]he wide variety of stem-formations implies that these morphologisations took place repeatedly over an extended period of time."

Indeed, the general pattern for the "verbalization" of **ĝʰoh₁* may be similar, *mutatis mutandis*, to that of **pér* 'through' > (verbal root) **per-* 'go through'. Thus **ĝʰoh₁* 'back, beyond' could have developed as a reduplicated intransitive verb of motion **ĝʰe-ĝʰóh₁-ti* ~ **ĝʰi-ĝʰéh₁-ti* 'goes back, goes beyond'; for the form and meaning, cf. **gᵘeh₂-* 'step' (Ved. *jígāti* etc., beside *e*-reduplication in the frozen participial form *jágat-* 'world').[22] This version, with generalized meaning 'go (away)', could survive directly in Germanic, while Gk. κιχη- could show the transitivization that might be expected if the sense 'go beyond' had survived more robustly than 'go back', whence 'reach (a goal, etc.)' (cf. Haudry 2005, §7 above). Indo-Iranian then emerges as showing a mix of innovation and archaism: here we find an oppositional middle, with innovative zero-grade beside the original active, but with archaic intransitive semantics: as LIV notes (s.v. ?**ĝʰeH-*, n.3), *jíhīte* 'moves' is mainly attested with preverbs; thus *+ápa* 'withdraws' may continue 'go back', while 'go beyond' may be continued in *+úd* 'rises', *+ní* 'sinks', or one may think in terms of the generalized 'go (away)' of Germanic.[23] Ved. *jáhāti*/Av. *zazāiti* 'leaves behind', in contrast, shows innovative transitive semantics (in effect, an "oppositional active" or "factitive" to the original intransitive sense; cf. Ved. *píparti* 'brings across' vis-à-vis Goth. *faran* 'go, travel'). This interpretation allows us to dispense with Hoffmann's somewhat awkward "intransitivization" of 'leaves behind' (via a suppressed direct object 'opponents') for the Avestan active forms that mean 'win': this usage could simply survive, in these specialized contexts, as an archaism, based on the original **ĝʰe-ĝʰóh₁-ti* 'goes beyond'.

But other versions of the verbalization process can easily be entertained, either as alternatives or as supplements. For example, **ĝʰoh₁* could have led, in the first instance (and by virtue of its *o*-vocalism), to an *o*-grade present (3sg.) **ĝʰóh₁-e*, (3pl.) **ĝʰéh₁-r̥(s)* 'goes back/beyond', precisely of the type Goth. *faran* (cf. Jasanoff 2003:76 for *faran* specifically, and also on the prominence of verbs of motion in this category generally).[24] In this event, Gmc.

[22] For the semantics ('back' >> 'go back'), cf. Hitt. *āppa* 'back, after', HLuv. *a-pi* 'afterward, further, again' ~ Hitt. *āppai-* 'go back; be finished' (Melchert [forthcoming]).

[23] Here it is worth remarking that *RV* VII.71.1a (cited also by Haudry, 2005:171) shows *ápa ... jihīte* 'withdraws' with subject *nág* 'night' and complement *uṣáso* 'dawn' (*ápa svásur uṣáso nág jihīte*, 'Night withdraws from her sister Dawn'), a lexical collocation not without interest for **ĝʰh₁-di̯-és*.

[24] As Melchert notes ([forthcoming]), the choice of the recessive *ḫi*-conjugation for secondarily inflected particles (as in Hitt. *āppai-* 'go back', *parā-* 'appear') seems surprising. One wonders whether, in addition to the factors Melchert considers, the *o*-vocalism of the derivational basis (at least at the Proto-Anatolian stage: HLuv. *a-pi* < **ópi*, Hitt. *parā* < **pró*) may have contributed to this choice as well.

*$g\bar{e}$- need not be dereduplicated, but could simply continue the weak stem, while Indo-Iranian and Greek could show a secondary reduplication related to the transitivization (cf. again Ved. *píparti*). Or else, in a development possibly also encouraged by the *o*-vocalism of *$\hat{g}^h oh_1$, one could imagine a stative perfect *$\hat{g}^h e$-$\hat{g}^h óh_1$-*e* 'is gone (to the back)' or 'is beyond', which might survive essentially intact (as 'is beyond', or with resultative sense 'has gone beyond') in the isolated Avestan perf.ptcple. *zazuuāh-/zazuš-* 'victorious' (whence secondary superlative *zazuštəma-*), which has the appearance of an archaism.[25] In the end, we must reckon with what Melchert ([forthcoming]) calls "the unpredictability of the transitivity of such formations from adverbs"—a feature that in itself may help support the present conception of this disparate verbal material as ultimately based on an adverbial particle. It is evident, finally, that whichever scenario(s) one prefers for the verbalization of *$\hat{g}^h oh_1$, a verbal root "*$\hat{g}^h eh_1$-" must have acquired a certain status at a relatively early period, to judge from primary derivatives; yet these are extremely rare, and perhaps actually restricted to an adjective *$\hat{g}^h eh_1$-*ro*- '(left) behind'[26] (Gk. χήρα 'widow', χῆρος 'bereaved', note also Dunkel 1987 on Lat. *hērēs* 'heir' and Gk. χηρωσταί 'plunderers of inheritance').[27]

All of this, to be sure, leaves Arm. *gam* 'come' unexplained. Perhaps Klingenschmitt was right to set up a root *$g^h eh_1$- 'reach, come to', to which Gk. κιχη- could also be assigned. But the last word may not have been said about the background of Arm. *gam*, the aorist of which is suppletive, based on two different PIE roots (*$g^{\mu} eh_2$- 'step' and *$g^{\mu} em$- 'come'). Klingenschmitt's rebuttal of the traditional etymology, based on an otherwise nonexistent unenlarged "*μeh_2-" (cf. *$\mu eh_2 d^h$- 'step' in Latin and Germanic), is convincing (1982:86); but other sources may be available, e.g. *μeh_2- 'turn' (Hitt. *wēḫzi*, weak stem *waḫ*-). The development of verbs meaning 'turn' to simple verbs of motion is a commonplace (cf. Eng. *wend, went*), and a development to 'come' (e.g. via 'turn to' or 'return') is not unthinkable. But such possibilities must be left for another day.[28]

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[25] The view that this form is neologistic (e.g. Kümmel 2000:673) is merely a consequence of the assumption that this meaning is innovatory for pres.act. *zazāiti*, which need not be the case.

[26] As I suggest elsewhere (Vine 2002:335-36), the adjective itself may have been back-formed from an earlier collective.

[27] Of the other primary-looking material often assigned to this root, generally with a sense 'empty, be lacking' (e.g. IEW 418-19), at least Gk. χατέω 'long for' (cf. χῆτος 'want, need') surely does not belong here, for phonological reasons (see LIV s.v. *$\hat{g}^h eh_1$-, n.1), and doubts may be entertained about the other material (Goth. *gaidw* 'lack', etc.) on other grounds.

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